

SAWYER COUNTY AP, WI, USA

WMO: 726508

Lat: 46.026N

Lon: 91.444W

Elev: 367

StdP: 96.99

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.1	-25.1	-32.1	0.2	-27.7	-29.0	0.3	-23.9	8.0	-5.5	7.1	-5.3	0.3	200	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	31.1	21.7	29.0	20.6	27.6	20.0	23.5	28.5	22.5	27.2	21.4	25.8	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.6	26.2	21.0	16.4	25.3	19.7	15.1	24.5	72.0	28.4	67.9	27.0	63.8	25.6	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.7	5.6	DB	-32.6	33.5	2.2	2.3	-34.1	35.2	-35.4	36.5	-36.6	37.8	-38.1	39.4	(4)
(5)			WB	-32.7	25.4	2.1	1.3	-34.2	26.4	-35.4	27.1	-36.6	27.9	-38.1	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-10.7	-8.8	-2.0	5.4	12.1	17.2	19.9	18.4	14.4	7.4	0.0	-7.6	(6)	
	DBStd	12.05	7.43	7.22	7.10	5.48	4.98	4.01	3.55	3.41	4.91	5.07	5.69	6.92	(7)	
	HDD10.0	2712	641	526	375	157	33	2	0	0	14	115	303	546	(8)	
	HDD18.3	4844	899	759	629	388	203	66	24	41	137	342	551	804	(9)	
	CDD10.0	1092	0	0	4	21	98	219	307	262	146	33	2	0	(10)	
	CDD18.3	182	0	0	0	1	11	33	73	44	19	2	0	0	(11)	
	CDH23.3	2097	0	0	1	13	149	403	857	474	182	18	0	0	(12)	
	CDH26.7	565	0	0	0	2	41	99	273	111	38	1	0	0	(13)	
(14) Wind	WSAvg	2.3	2.1	2.2	2.4	2.8	2.7	2.3	2.0	1.9	2.2	2.3	2.3	2.1	(14)	
(15) Precipitation	PrecAvg	834	24	25	38	68	92	107	120	110	96	82	47	30	(15)	
	PrecMax	1193	38	53	92	186	135	174	264	194	176	161	128	53	(16)	
	PrecMin	573	5	6	14	23	31	39	46	29	25	26	10	7	(17)	
	PrecStd	150	9	13	19	39	29	35	55	50	40	41	29	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	10.2	19.6	25.2	30.6	31.7	34.0	32.1	30.1	26.0	17.5	8.1	(19)	
		MCWB	2.5	6.3	13.7	14.7	19.4	22.1	22.9	22.7	21.5	18.1	11.5	5.4	(20)	
	2%	DB	3.5	6.5	15.2	21.9	27.3	29.1	31.7	29.2	27.5	22.2	13.9	5.3	(21)	
		MCWB	1.2	3.6	10.4	12.1	17.5	20.3	22.0	20.9	20.1	15.5	9.9	3.0	(22)	
	5%	DB	1.5	3.5	11.5	18.8	24.2	27.5	29.4	27.6	25.5	18.9	11.3	3.0	(23)	
		MCWB	0.2	1.2	7.3	10.2	15.9	19.4	21.1	20.2	19.1	13.5	7.4	1.4	(24)	
	10%	DB	-0.4	1.3	8.0	16.0	22.0	25.9	27.7	26.1	23.1	16.3	8.7	1.7	(25)	
		MCWB	-1.7	-0.5	4.5	8.8	14.1	18.2	20.6	19.3	17.6	11.9	5.6	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	6.6	14.8	16.2	22.0	24.1	25.5	24.3	23.1	20.5	13.2	6.8	(27)	
		MCDB	5.5	9.0	18.3	22.6	27.3	29.4	30.7	28.8	27.2	23.9	15.7	7.8	(28)	
	2%	WB	1.6	3.9	10.7	13.2	19.1	22.3	23.8	23.0	21.5	16.5	10.3	3.4	(29)	
		MCDB	3.0	6.4	13.5	19.5	24.7	27.4	29.0	27.4	26.1	20.7	13.8	4.8	(30)	
	5%	WB	0.2	1.4	7.6	11.1	17.2	20.7	22.8	21.7	20.1	14.4	7.8	1.7	(31)	
		MCDB	1.6	3.4	11.3	17.3	22.3	25.3	27.7	25.6	24.1	18.1	10.6	3.0	(32)	
	10%	WB	-1.6	-0.3	4.7	9.3	15.5	19.4	21.6	20.5	18.7	12.3	5.8	0.4	(33)	
		MCDB	-0.5	1.1	8.0	15.6	20.4	23.7	26.1	24.5	22.5	15.9	8.5	1.7	(34)	
(35) Mean Daily Temperature Range	MDBR		10.3	11.8	12.0	12.8	13.9	13.1	13.3	13.1	12.8	10.9	9.0	8.9	(35)	
	5% DB	MCDBR	10.2	12.1	14.6	18.4	17.2	15.1	15.0	14.7	14.3	15.0	13.5	9.6	(36)	
		MCWBR	8.5	9.3	9.6	10.4	8.8	7.9	7.3	7.8	8.0	9.3	9.4	7.9	(37)	
	5% WB	MCDBR	9.6	10.4	13.3	15.8	13.9	12.4	12.6	12.3	11.9	12.3	12.1	8.6	(38)	
		MCWBR	8.1	8.1	9.2	9.9	7.9	7.4	7.1	7.4	7.3	8.4	9.0	7.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.247	0.274	0.308	0.344	0.375	0.374	0.369	0.377	0.358	0.312	0.286	0.263	(40)	
	taud		2.405	2.332	2.369	2.401	2.350	2.376	2.418	2.408	2.444	2.560	2.523	2.430	(41)	
	Ebn at Noon		878	910	924	914	892	892	891	871	859	861	823	814	(42)	
	Edh at Noon		69	91	103	110	120	118	112	109	96	74	63	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.47	2.56	3.72	4.52	5.31	5.81	6.08	5.08	3.90	2.41	1.43	1.02	(44)	
	RadStd		0.15	0.23	0.34	0.47	0.40	0.43	0.24	0.34	0.24	0.28	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page